

Homework 4: Stability Analysis: The Trace-Determinant Plane

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BCS A, E, F, G

Consider an ODE system described by the matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

where the trace ($\text{tr } A$) of A is defined as $\text{tr } A = a + d$ and the determinant ($\det A$) of A is given by $\det A = ad - bc$. The eigenvalues (λ) of A are determined from the characteristic polynomial $\lambda^2 - (\text{tr } A)\lambda + \det A = 0$ of A using the quadratic formula:

$$\lambda = \frac{\text{tr } A \pm \sqrt{\text{tr}^2 A - 4 \det A}}{2} = \frac{\text{tr } A}{2} \pm \frac{\sqrt{\text{tr}^2 A - 4 \det A}}{2}.$$

Discuss the qualitative characteristics of the phase portrait for a two-dimensional system based on the trace and determinant of the 2×2 coefficient matrix A (The Trace-Determinant Plane). Give a pictorial summary by the end of discussion that projects the stability analysis in a nutshell.

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Good Luck